Measuring Validator Utility in Generate–Verify–Repair NetOps Pipelines: a Confirmatory Paired Study with Shared Initial Candidates

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***Abstract*—We preregistered and executed a paired confirmatory study (AC-2026-03) to evaluate whether a multidimensional validator metric suite predicts end-to-end agentic NetOps success better than a single verifier pass rate. Using a synthetic intent→configuration generator and a hosted generate–validate–repair (GVR) adapter under shared_initial_candidate semantics, we ran 200 paired tasks (one shared initial generation per task reused for baseline and guarded). The deterministic validator (deterministic_netops_validator_v5, v5.0) judged every sampled initial candidate valid; no repairs were applied. Baseline = 200/200, guarded = 200/200, paired difference = 0.00 (bootstrap 95% CI [0.00, 0.00], exact McNemar two-sided p = 1.0; bootstrap_seed = 1729, resamples = 10000). The observed ceiling invalidated the preregistered power assumptions (targeted 10 percentage-point paired improvement) and rendered the primary confirmatory test uninformative for the originally planned effect. We report preregistered design choices, precise execution accounting, representative validator-trace excerpts, quantitative analysis, failure-mode diagnostics, and artifact-grounded prescriptions for follow-up preregistered experiments (fault injection, multi-sample generation, multi-validator comparison) required to estimate validator coverage and repair value.**

I. Introduction

Automating NetOps workflows with generative models requires reliable mechanisms to avoid harmful, silent, or wrong-but-plausible actions. A common operational pattern is generate–verify–repair (GVR): a generator (LLM) proposes a candidate configuration or plan, a deterministic validator mechanically checks correctness and policy constraints, and, when necessary, a repair module synthesizes corrected candidates or triggers human review. Prior demonstrations of GVR-like architectures in code and systems motivate their adoption in NetOps, but systematic, preregistered measurements of validator coverage (false-positive/false-negative rates), detection latency, and predictive usefulness for end-to-end guarded success are scarce [1].

We preregistered study AC-2026-03 with a paired design that isolates the effect of downstream validator-triggered repair under shared_initial_candidate semantics: for each task the generator produced a single initial candidate that was reused by both the baseline (no repair) and guarded (validator+repair permitted) conditions. Under these semantics any paired difference can only arise downstream from validator-triggered repair, not from independent generator sampling. The primary estimand was the paired_success_rate_difference_guarded_minus_baseline and the preregistered confirmatory hypothesis (H1) expected that a multidimensional validator-metric suite would explain more variance in end-to-end success than a single verifier pass rate. Our main contribution is a fully preregistered, artifact-archived experiment and an exact, transparent report of a degenerate confirmatory outcome (ceiling) with concrete, artifact-grounded prescriptions for follow-up designs that can measure validator utility.

II. Related Work

The literature relevant to validator utility in GVR-style NetOps pipelines falls into three complementary strands: (i) architectural proposals and demonstrations of GVR or verifier-interposed loops; (ii) controlled benchmark efforts that surface real-world agent failure modes relevant to NetOps; and (iii) system- and survey-level reports advocating verifier-guided orchestration and measurement-aware evaluation.

Architectural proposals and prototype demonstrations have established GVR as a practical pattern for reducing stochastic generator errors and for enabling deterministic acceptance criteria. Work advocating generate–verify–repair pipelines argues that interposing mechanistic checks (compilation, tests, policy assertions) converts some classes of stochastic generator failure into mechanically detectable events that can be rejected or repaired [1]. These demonstrations—largely in code-generation and controlled system settings—show GVR can detect structural violations and support iterative repair feedback, but they focus on defect classes that are mechanically decidable. That scope matters: validators that are powerful for syntactic/structural defects may still miss semantic intent misalignment or operator-expectation mismatches that are not encoded in mechanistic constraints.

Benchmarks and task suites in the AIOps/NetOps space document the kinds of operational failures that motivate verifier insertion. Recent benchmark work emphasizes ticket noise, false premises, and wrong-but-plausible outputs that complicate end-to-end automation; controlled suites have been used to measure agent brittleness and the practical costs of erroneous tool calls [2],[3]. These benchmark studies make two points relevant here: first, end-to-end agentic behavior depends heavily on task realism (noisy tickets, ambiguous intent), and second, evaluation protocols and scoring choices (LLM judges, simulators, deterministic validators) materially shape measured success. Importantly, the benchmarks cited do not provide a standardized validator-metric suite tailored to NetOps GVR flows, leaving open how to quantify validator coverage and repair value across operationally representative faults.

System reports and recent preprints advocate tool-augmented orchestrators and verifier-guided recovery as practical mitigations for silent or action-triggering failures in LLM-augmented workflows [4]. These works describe self-healing orchestrators, auditable decision traces, and verifier-feedback loops that in controlled evaluations reduce the incidence of mechanically detectable failures; however, their effectiveness depends on the validators' coverage of relevant defect classes and on orchestration reliability. Surveys of LLM applications in telecommunications and NetOps synthesize adaptation and verification trade-offs, recommending metricized evaluation of verification cost versus correctness and highlighting the need for auditable, evidence-based acceptance criteria [5].

Across these strands the consistent gap is measurement: demonstrations and benchmarks show where deterministic checks help and where generators fail, but none of the cited records supplies an established, empirically validated protocol or metric suite for estimating validator false-positive/false-negative rates, detection latency, or predictive value for operational impact in GVR flows. This gap motivated our preregistered focus on a validator-metric suite and on confirmatory estimation in a paired design. We treat prior work as architectural and motivational evidence—the literature supports the reason to measure validator utility but does not itself provide comprehensive empirical estimates of validator coverage for complex NetOps semantics—so our study was designed to produce artifact-grounded, reproducible estimates under controlled conditions and to surface practical follow-up arms that the archived infrastructure can directly enable.

III. Methodology

Preregistration and primary design. The study preregistration (AC-2026-03) specified a paired confirmatory design using the hosted_netops_gvr_v1 adapter family with generation_semantics = shared_initial_candidate and initial_generation_calls_per_task = 1. Planned task_count = 200 pairs (planned_episode_count = 400), planned maximum_model_calls = 400, and model_scope = gpt-5-mini via the provider bundle recorded as 'openai_responses'. The preregistered primary estimand was paired_success_rate_difference_guarded_minus_baseline; the primary test was an exact two-sided McNemar with paired bootstrap percentile confidence intervals (bootstrap_resamples = 10000, bootstrap_seed = 1729). Targeted detectable effect was an absolute 0.10 paired improvement with α = 0.05 and ~80% power under conservative planning assumptions.

Generator, tasks, and constraints. Tasks were drawn deterministically from deterministic_netops_task_generator_v5 (task_family = intent_configuration_repair_v1). Each task encodes: an initial_state, an expected_final_state, an explicit intent string, allowed_fields/allowed_touched_fields, workflow_policies (e.g., minimum_active_in_groups, must_be_down_for_fields), and a required_command_sequence. Task selection followed a preregistered deterministic index rule (challenging_workflow_stress_v1). Contamination/exclusion rules (duplicate SHA-256 and embedding-similarity > 0.95) were preregistered; the execution/analysis artifacts record that no contamination exclusions occurred (analysis/contamination_summary.csv empty).

Model and sampling configuration. The declared model version in execution/model_configuration.json is "gpt-5-mini"; execution accounting records model_calls_used = 200 (one shared initial generation per pair) and maximum_model_calls = 400 as preregistered. Important artifact-grounded note: execution/model_configuration.json records temperature = null (temperature = null/unset). Null/unset is explicitly not evidence of deterministic provider-side sampling and does not imply temperature = 0; provider sampling variability remains possible and is recorded in provider-call audit fields. Provider-call audit in deterministic_reconciliation reports hosted_model_call_event_count = 200 and stage_counts.shared_initial_generation = 200, consistent with the preregistered single shared generation per task.

Validator and scoring. The deterministic validator used was deterministic_netops_validator_v5 (validator_version = 5.0). The validator implements mechanistic intent-constraint checks and emits per-episode fields: normalized_configuration, parsed_command_count, required_command_count, observed_touched_field_count, violation_count/violation_codes, valid (boolean), repair_applied (boolean), and repair artifacts when applicable. The preregistered binary scoring rule (full_intent_constraint_validation) uses validator.valid as the success label for the confirmatory estimand. The pipeline permitted at most one repair call when validator rejected a candidate; repair events would be recorded with repair_applied = true and repair_provider_trace fields.

Execution, exclusions, and missingness. The preregistered maximum_attempts_per_call = 1 (no manual retries). The execution manifest reports completed_episode_count = 400, failed_episode_count = 0, and complete_pair_count = 200. The preregistered missingness and contamination plans were followed: confirmatory analyses used complete pairs only; artifacts include analysis/missingness_summary.csv and analysis/contamination_summary.csv. All raw responses, per-episode validator traces, provider-call traces, and analysis artifacts are archived in the execution bundle and referenced in this paper.

IV. RESULTS

Execution accounting and primary outcomes. The adapter completed the run exactly as preregistered: planned_episode_count = 400, completed_episode_count = 400, failed_episode_count = 0, and model_calls_used = 200 (one shared initial generation per pair). Provider-call audit: hosted_model_call_event_count = 200, completed_hosted_model_call_event_count = 200, cache_hit_count = 0, cache_miss_count = 200, stage_counts.shared_initial_generation = 200. No contamination exclusions or near-duplicate exclusions occurred.

Primary confirmatory result. Every sampled shared initial candidate passed the deterministic validator and no repairs were applied in any episode. Analysis artifacts report baseline_success_count = 200/200 (success_rate = 1.0) and guarded_success_count = 200/200 (success_rate = 1.0). The paired contingency (analysis/paired_contingency_table.csv) is n_11 = 200, n_10 = 0, n_01 = 0, n_00 = 0. The paired_success_rate_difference_guarded_minus_baseline = 0.00. Exact McNemar two-sided test p-value = 1.0. Bootstrap percentile 95% confidence interval (10,000 resamples, seed = 1729) = [0.00, 0.00]. The numeric artifacts (analysis/condition_summary.csv, analysis/paired_contingency_table.csv, analysis/analysis_log.jsonl) match these reported values.

Representative validator-trace excerpt (artifact-grounded). To illustrate validator outputs and why no repairs were triggered, we include verbatim key fields from execution/scoring/task-000004-guarded.json (pair-000004):

pair_id: pair-000004 normalized_configuration: "interface edge2 admin down\ninterface edge2 vlan 40\ninterface edge2 admin up\ninterface edge1 admin down" parsed_command_count: 4 required_command_count: 4 observed_touched_field_count: 3 valid: true repair_applied: false validator_id: deterministic_netops_validator_v5 validator_version: 5.0

Equivalent validator_trace.validation_after.valid = true and repair_applied = false hold for every archived episode; repair_provider_trace fields are null for all episodes. Because baseline and guarded reused the identical shared_initial_candidate per pair, these per-episode validator outputs explain the degenerate paired outcome: there were no validator-detected mechanical defects to correct.

Secondary and exploratory diagnostics. Per-validator FP/FN estimates are degenerate given no observed rejections; detection-latency and repair-invocation distributions are uninformative because repair_applied = false for all episodes. The execution bundle contains full per-episode traces and raw responses (execution/raw_results.jsonl) to support reanalysis under alternative scoring or injected-fault conditions. Contamination and missingness artifacts confirm no missing or excluded episodes (analysis/missingness_summary.csv: complete_pairs = 200).

Artifact-grounded diagnostic interpretation. Under shared_initial_candidate semantics any paired difference could only arise from validator-triggered repair. The degenerate all-valid outcome is explained by three artifact-grounded determinants visible in the archive: (1) the deterministic task generator produced candidates that satisfied the validator's mechanistic constraints for the recorded generation seeds; (2) the single-sample-per-task sampling regime removed generator variability as a source of disagreements; (3) mechanistic validators detect structural and policy violations but can be blind to semantic intent misalignment (correct-by-structure yet incorrect-by-intent). These jointly explain why no repairs occurred and the confirmatory test could not detect the preregistered 10-percentage-point improvement target.

V. DISCUSSION

Design implications of shared_initial_candidate semantics. The preregistered shared_initial_candidate choice isolates downstream validator effects by ensuring both conditions evaluate the same initial candidate for each pair. This isolation is scientifically valuable because any observed paired difference must derive from validator-triggered repair. It also concentrates the risk of a single-sample ceiling: if the generator sample aligns with the validator for all tasks, the design yields zero observed variance in the primary estimand, as occurred here. We emphasize this property because it determines what the paired estimator can and cannot measure: it cannot detect differences attributable to generator sampling or to alternative prompt constraints when the sampling semantics were shared.

Operational interpretation and validator scope. The deterministic validator (deterministic_netops_validator_v5) reliably enforces mechanistic intent-constraint checks (parsed_command_count, required_command_count, violation_codes, valid flag). However, artifact-grounded evidence here shows that correctness-by-structure does not imply semantic intent alignment under operator expectations. That limitation is intrinsic to validators that lack an external intent model or operator feedback; detecting semantic mismatches requires either richer intent models, human-in-the-loop signals, or tailored semantic validators. The present execution does not imply validators are unnecessary—rather, it demonstrates that in a synthetic bank and single-sample regime the validator had nothing mechanical to correct.

Expanded diagnostics grounded in execution artifacts. The archived execution provides explicit numeric diagnostics that clarify why the confirmatory estimator was uninformative: model_calls_used = 200 (one shared generation per pair), complete_pair_count = 200, baseline_success_count = 200, guarded_success_count = 200, and repair_applied = false observed for every episode. The paired contingency is a point mass (n_11 = 200), and the bootstrap procedure (bootstrap_seed = 1729, resamples = 10000) produced a degenerate sampling distribution with 95% percentile CI [0.00, 0.00]. These exact values are available in the analysis artifacts (condition summary and paired contingency outputs) and show that the confirmatory null is not a statistical fluke but a deterministic consequence of the joint generator–validator–sampling configuration used.

Practical lessons for preregistered validator evaluation. From a methodological perspective, two trade-offs are clear and artifact-evident. First, shared_initial_candidate is a powerful control: it guarantees that any observed paired improvement is attributable to downstream validator/repair logic alone. Second, that control increases sensitivity to ceiling effects when generation produces validator-compliant candidates for the chosen seeds. For studies whose primary aim is to measure validator coverage (FP/FN) and repair utility, the preregistered design should therefore include one or both of the following preregistered modifications to avoid degeneracy: (a) a controlled fault-injection arm that deterministically introduces a known fraction f of invalid cases (syntactic, configuration-logic, or semantic-intent swaps) into the evaluation set so that the validator will face realistic defects; (b) a multi-sample generation arm with initial_generation_calls_per_task >= 3 independent draws per task to reintroduce generator variability and produce candidate diversity that can trigger validator rejections and repairs. Both prescriptions were articulated in the archived post-run prescriptions and are directly implementable within the same adapter semantics.

What validator metrics become estimable under those modifications. If faults are injected or multiple independent draws are allowed, the validator-metric suite preregistered in AC-2026-03 becomes estimable: (i) per-validator false-positive rate (FP-rate) as the fraction of validator.accept = true when the task's ground-truth is invalid; (ii) per-validator false-negative rate (FN-rate) as validator.reject = false for candidates that do actually violate the required intent sequence; (iii) mean detection latency measured from model-response receipt to validator decision (records include timestamps in provider- and validator-traces); (iv) repair-invocation rate and post-repair success fraction when repair_applied = true is observed. These metrics map directly to the fields already emitted by the validator in the archived traces (valid, violation_count, repair_applied, parsed_command_count) and to provider-call audit counters; no additional opaque instrumentation is required to compute them once non-degenerate data are present.

Reproducibility and artifact-grounded protocol refinements. The execution artifacts document concrete accounting fields that support exact replication of the intended follow-ups: provider-call audit counts (hosted_model_call_event_count = 200), the model identifier (declared_model_version = gpt-5-mini), the recorded temperature = null/unset (explicitly not evidence of deterministic sampling), and the validator identifier/version (deterministic_netops_validator_v5, v5.0). A follow-up preregistration should (and can, per the archived infrastructure) explicitly set: (1) initial_generation_calls_per_task = k (e.g., k = 3) with independent generation seeds recorded for each draw; (2) a deterministic fault-injection schedule specifying fault classes and the exact fraction f of tasks affected; and (3) a plan to measure detection latency using the existing timestamps. These concrete protocol elements are already supported by the adapter and manifest fields and would remove ambiguity about sampling and allow FP/FN estimation without inventing new instrumentation.

Threats to validity revisited with artifact evidence. Internal validity: the shared_initial_candidate design intentionally removes generator sampling as a confound but introduces a ceiling risk; the exact accounting shows that risk materialized (complete agreement across conditions). Construct validity: validator.valid captures mechanistic constraint satisfaction but not semantic intent alignment—this remains the central construct gap and explains why correct-by-structure outputs can still be operationally unacceptable. External validity: the study executed one model (gpt-5-mini) and one deterministic validator against a synthetic task bank; therefore caution is required when extrapolating to multi-vendor production networks or to models and validators with different coverage. Conclusion validity: the archived bootstrap and contingency outputs (seed/resamples specified) demonstrate the degenerate inferential state (zero variance) and justify treating the confirmatory hypothesis as untestable under these run parameters rather than as a failed hypothesis test.

Operational implications and recommended reporting practices. For experimenters and operators evaluating GVR pipelines, we recommend preregistering both (a) the sampling semantics (shared_initial_candidate vs independent draws) and (b) explicit anti-ceiling design elements (fault injection or multi-draw sampling) whenever the evaluation goal is to estimate validator coverage or repair utility. Empirically reporting the exact counts we used (model_calls_used, complete_pair_count, baseline/guarded success counts, repair_applied counts, bootstrap_seed/resamples) increases interpretability because readers can immediately diagnose whether a noninformative null arose from lack of validator failures or from insufficient sample diversity.

Summary. The execution artifacts and numeric diagnostics make the outcome clear: under the chosen shared_initial_candidate semantics, the combination of deterministic seeds, single-sample generation, and a mechanistic validator that checks structural constraints produced a ceiling effect (200/200 valid). That outcome is scientifically informative about the joint behavior of generator+validator under these settings and usefully constrains what conclusions can and cannot be drawn. It also supplies a concrete, artifact-grounded roadmap—fault injection, multi-sample generation, and multi-validator comparison—that will enable future preregistered studies to estimate the validator metrics of interest and to quantify repair utility in operationally meaningful conditions.

VI. Limitations

- Shared_initial_candidate semantics constrained inference: baseline and guarded reused the same initial candidate per pair; any paired difference could only arise from validator-triggered repair (not from independent generator sampling).
- Synthetic task bank (difficulty 'very_hard'/'extreme') is not equivalent to heterogeneous production networks (multi-vendor devices, real ticket noise); external validity is limited.

- Single-model experiment: only gpt-5-mini was executed; no multi-model generality claims are made.
- No repairs were applied because all initial candidates passed the deterministic validator; therefore the study could not estimate repair effectiveness or validator false-negative behavior in this execution.
- Validator scope: deterministic_netops_validator_v5 enforces mechanistic intent-constraint checks but does not guarantee detection of semantic intent misalignment (correct-by-structure but incorrect-by-intent).
- Recorded execution/model_configuration.json temperature = null/unset does not imply deterministic sampling; provider-side sampling behavior may vary and is documented in execution manifests.

VII. Conclusion

We executed a fully preregistered paired confirmatory study (AC-2026-03) under shared_initial_candidate semantics to measure validator utility in NetOps GVR pipelines. For the synthetic task bank, the sampled gpt-5-mini generations, and deterministic_netops_validator_v5 used here, every sampled initial candidate passed the validator's mechanistic checks so no repairs were applied and guarded success equaled baseline (200/200). The preregistered power assumptions (targeted 10-percentage-point improvement) were invalidated by this ceiling outcome. The result is artifact-grounded and reproducible from the archived bundle. We publish the exact execution and analysis artifacts to support replication and provide concrete, preregistered experiment prescriptions (fault injection, multi-sample generation, multi-validator comparison) that are required to produce estimable validator FP/FN behavior and repair value in operationally meaningful conditions.

Disclosure Statement

Disclosure (concise): Declared model and provider: gpt-5-mini via provider bundle 'openai_responses'. Orchestration/adapter: hosted_netops_gvr_v1. Deterministic validator: deterministic_netops_validator_v5 (validator_version = 5.0). Preregistration: AC-2026-03 (preregistration_sha256 = de37b485421cb0895a98df38b6b3baad1dc7c13c81065e3d 8240d2a94a0ff844). Master prompt immutable reference: provenance/master_prompt.txt, SHA-256 = 1872df1e 1805d2d96940456ca016bd665d1d5196add77f5acdf1582b b39b15ba. Autonomy boundary: a human Research Director selected the broad topic family and approved the master prompt before the autonomy lock; after the lock the AI system autonomously performed literature review, preregistration, experiment design, execution, analysis, internal peer review and manuscript drafting without manual editing of technical content. Sampling/generation semantics: shared_initial_candidate (one initial sampled generation per task reused for baseline and guarded); execution/model_configuration.json recorded temperature = null/unset (null/unset is not evidence of deterministic sampling and does not imply temperature = 0). Call budget and usage (preregistered): maximum_model_calls = 400; actual model_calls_used = 200 (one shared initial generation per pair). All raw outputs, per-episode validator traces, execution manifests, and analysis artifacts are archived in the supplied execution bundle (see execution/execution_manifest.json). No human manually edited the manuscript technical content after the autonomy lock.

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